

第一次 (自学部分, 奖励作业题, 5分) } P₁

证明: Carathéodory's criterion: (or extension Thm).

Assume that $\mathcal{E}$ is a semiring on Ω . $\mu(\cdot): \mathcal{E} \rightarrow [0, \infty)$ is a additive function: i) $\mu(\emptyset) = 0$, ii) $\forall A_n \in \mathcal{E}, \sum_{n=1}^{\infty} A_n \in \mathcal{E}$,
$$\mu\left(\sum_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

Then (1) μ has an extension measure $\bar{\mu}$ on $(\Omega, \sigma(\mathcal{E}))$.
s.t. $\bar{\mu}|_{\mathcal{E}} = \mu$.

(2) If μ is σ -finite function: $\Omega = \sum_{n=1}^{\infty} A_n, A_n \in \mathcal{E}$,
 $\mu(A_n) < +\infty, \forall n \geq 1$, then $\bar{\mu}$ is the unique
extension.

证明上述定理的提示及参考书 (第7周交)

Step 1: 证明下面引理1.

Lemma 1. If $\mu^*(A) \triangleq \inf \left\{ \sum_{n=1}^{\infty} \mu(A_n) : A_n \in \mathcal{E}, A \subseteq \bigcup_n A_n \right\}, \forall A \subseteq \Omega$.

then (i) $\mu^*(\emptyset) = 0$; (ii) $A_1 \subseteq A_2 \Rightarrow \mu^*(A_1) \leq \mu^*(A_2)$;

(iii) $\mu^*\left(\bigcup_n A_n\right) \leq \sum_{n=1}^{\infty} \mu^*(A_n), \forall A_n \subseteq \Omega$;

(iv) $\mu^*|_{\mathcal{E}} = \mu$. i.e. μ^* is an 外测度 on $(\Omega, 2^\Omega)$, P377-378
P377-378

(该引理的证明, 见英文教材: P377, Theorem 1. ($\mathcal{E} = \mathcal{G}_0$))

Step 2. 证明下面引理2:

(见英文教材: P378, Thm 2)

Lemma 2. Define $\mathcal{F}^* = \{A \in \Omega : \mu^*(B) = \mu^*(BA) + \mu^*(BA^c), B \in \mathcal{F}\}$ 12

Then (1) $\mathcal{F}^*$ is a σ -algebra on Ω ; (或中财大 P20, Thm 1.4.1) 英文: P278-380

(2) $\mathcal{E} \subseteq \mathcal{F}^* \Rightarrow \sigma(\mathcal{E}) \subseteq \mathcal{F}^*$;

(3) μ^* is a measure on $\mathcal{F}^*$.

step 3. 证明下述引理 3. (or 中财大 P21)

Lemma 3. If μ_1 and μ_2 are two finite measure

on $\sigma(\mathcal{E})$, $\mathcal{E}$ is π -system, $\mu_1|_{\mathcal{E}} = \mu_2|_{\mathcal{E}}$, then

$\mu_1 = \mu_2$. (P381, Thm 3, 课堂讲过). (λ-π 定理 or MCT)

Proof of Carathéodory's criterion:

(A) Using (2) of Lemma 2 (2), $\sigma(\mathcal{E}) \subseteq \mathcal{F}^*$. Define.

$\bar{\mu} = \mu^*|_{\sigma(\mathcal{E})}$. Then, based on Lemma 2 (3), $\bar{\mu}$ is a measure

on $(\Omega, \sigma(\mathcal{E}))$. Using Lemma 1 (iv), $\bar{\mu}|_{\mathcal{E}} = \mu$. Thus

$\bar{\mu}$ is an extension measure on $(\Omega, \sigma(\mathcal{E}))$ of μ .

(B) Uniqueness. Assume that $\mu_1|_{\mathcal{E}} = \mu_2|_{\mathcal{E}} = \mu$.

Define $\mathcal{G} = \{A \in \sigma(\mathcal{E}) : \mu_1(A \cap A_n) = \mu_2(A \cap A_n)\}$, $n \geq 1$.

Based on Lemma 3, $\mu_1|_{A_n \cap \sigma(\mathcal{E})} = \mu_2|_{A_n \cap \sigma(\mathcal{E})}$.

As $\Omega = \bigcup_{n=1}^{\infty} A_n$, we have $\mu_1 = \mu_2$.

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